

DSC40B:
Theoretical Foundations of Data
Science II

Lecture 4: *Expected time complexity*

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Part A: Probabilistic analysis vs randomized algorithms



Previously

- ▶ **Worst-case and best-case time complexity analysis**
 - ▶ Worst-case time complexity analysis
 - ▶ Most commonly used. Guarantees performance even in the worst case
- ▶ However, both worst- and best-case time can be caused by just some specific input
- ▶ How about average time complexity
 - ▶ Intuitively measures how the algorithm works on a typical input?



Expected Analysis

- ▶ **Probabilistic method:**
 - ▶ Given a distribution for all possible inputs
 - ▶ Derive expected time based on distribution
- ▶ **Randomized algorithm:**
 - ▶ Add randomness in the algorithm
 - ▶ Analyze the expected behavior of the algorithm



A simple example

```
def linear_search(A, k):  
    for i, x in enumerate(A):  
        if x == k:  
            return i  
    return None
```

- ▶ What is worst case time complexity?
- ▶ What is expected / average time complexity?



Expected Running Time

- ▶ Expected / average running time
 - ▶ $ET(n) = \sum_I \Pr(I) \text{time}(I)$
 - ▶ $\Pr(I)$ = probability of input type I
 - ▶ $\text{time}(I)$ = running time given input type I
- ▶ To analyze, **need to assume** a probabilistic distribution for all inputs



Linear-search algorithm

- ▶ Expected running time =

$$\Pr(K \notin A) \text{time}(K \notin A) + \sum_{i=1}^n \Pr(A[i] = K) \text{time}(A[i] = K)$$

- ▶ If we assume

- ▶ $\Pr(K \notin A) = 0$

- ▶ All permutations are equally likely

- ▶ implies $\Pr(A[i] = K) = \frac{1}{n}$

- ▶ $\text{time}(A[i] = K) = ci$

- ▶ Then expected running time = $\sum_i \left(\frac{1}{n}\right) * ci = \Theta(n)$



Strategy for finding **Average Case**

- ▶ Step 0: Make assumption about distribution of inputs
- ▶ Step 1: Determine the possible cases
- ▶ Step 2: Determine the probability of each case
- ▶ Step 3: Determine the time taken for each case
- ▶ Step 4: Compute the expected time (average) , which is the sum of time for each case weighted by its probability



Remark

- ▶ For probabilistic analysis

- ▶ An input probabilistic distribution input model **has to be assumed!**
- ▶ For a fixed input, the running time is fixed.
- ▶ The average / expected time complexity is for if we consider running it for a range of inputs, what the average behavior is.

- ▶ Randomized algorithm

- ▶ No assumption in input distribution!
- ▶ Randomness is added in the algorithm
 - ▶ For a fixed input, the running time is **NOT** fixed.
 - ▶ The expected time is what we can expect when we run the algorithm on **any single input**.



Part B: Analyzing randomized algorithms



Randomized linear search example

```
def rand_linear_search(A, k):  
    random.shuffle(A)  
    for i, x in enumerate(A):  
        if x == k:  
            return i  
    return None
```

- ▶ What is expected / average time complexity?
 - ▶ Assuming we only search for keys already in A
 - ▶ $\Pr(A[i] = k) = \frac{1}{n}$
 - ▶ $ET(n) = \Pr(k \notin A) \text{time}(K \notin A) + \sum_{i=1}^n \Pr(A[i] = k) \text{time}(A[i] = k)$
 - ▶ $= \sum_i \left(\frac{1}{n}\right) * ci = \Theta(n)$



Strategy for finding Expected Case for Randomized Alg

- ▶ ~~Step 0: Make assumption about distribution of inputs~~
- ▶ Step 0: Inspect the randomness in the algorithm
- ▶ Step 1: Determine the possible cases
- ▶ Step 2: Determine the probability of each case
- ▶ Step 3: Determine the time taken for each case
- ▶ Step 4: Compute the expected time (average) , which is the sum of time for each case weighted by its probability



Review of Expectation

- ▶ X is a random variable
- ▶ The expectation of X is
 - ▶ $E(X) = \sum_I \Pr(X = I) I$
 - ▶ E.g, coin flip
- ▶ Linearity of expectation:
 - ▶ $E(X_1 + X_2) = E(X_1) + E(X_2)$
- ▶ Conditional expectation:
 - ▶ $E(X) = E(X | Y) \Pr(Y) + E(X | \text{Not } Y)(1 - \Pr(Y))$



Typical type of randomness

- ▶ Random shuffle / permutation of n elements

- ▶ `random.shuffle(A)` :
- ▶ `x = np.random.randint(0, n)`; or `x = random(n)`
 - ▶ $\Pr[x = i] = \frac{1}{n}$ for any $x \in [0, n - 1]$

- ▶ Random coin flip

- ▶ Flip a coin and the probability of head is p . This means that $\Pr[tail] = 1 - p$
- ▶ For a “fair coin”, $p = 0.5$



Use of linearity of expectation

```
Input   : Array A of  $n$  integers.  
function func1(A[],  $n$ )  
1  $s \leftarrow 0$ ;  
2 for  $i \leftarrow 1$  to  $n$  do  
3   |  $A[i] \leftarrow A[n - i + 1]$ ;  
4   |  $s \leftarrow s + \text{func2}(A, n)$ ;  
5 end  
6 return ( $s$ );
```

$$ET_1(n) = n ET_2(n) + cn$$

- ▶ $ET_2(n)$ = expected running time for func2
- ▶ What is $ET_1(n)$?



Use of Conditional expectation

```
function func1(A[],n)
1 Flip a coin;
2 if heads then
3   | a ← func2(A,n);
4 else
5   | a ← func3(A,n);
6 end
7 return (a);
```

$$ET_1(n) = \Pr(head) ET_2(n) + (1 - \Pr(head)) ET_3(n) + c$$

- ▶ $ET_2(n)$ = expected running time of func2
- ▶ $ET_3(n)$ = expected running time of func3
- ▶ What is the expected running time of func1?



Randomized example 1

```
Func1( $A, n$ )  
  /*  $A$  is an array of integers  
1  $s \leftarrow 0$ ;  
2  $k \leftarrow \text{Random}(n)$ ;  
3 for  $i \leftarrow 1$  to  $k$  do  
4   | for  $j \leftarrow 1$  to  $k$  do  
5   |   |  $s \leftarrow s + A[i] * A[j]$ ;  
6   | end  
7 end  
8 return ( $s$ );
```

► $\text{Random}(n)$:

- returns a number k s.t. the probability that $k = i$ for any $i \in [1, n]$ is $\Pr[k = i] = \frac{1}{n}$



Running time analysis

- ▶ **Worst Case:**

- ▶ $T(n) = \Theta(n^2)$

- ▶ **Expected running time:**

- ▶ Step 1: identify different possible cases
 - ▶ Step 2: find the probability of each case
 - ▶ Step 3: find the running time of each case

- ▶
$$\begin{aligned} ET(n) &= \sum_{i=1}^n \Pr(k = i) \cdot (ci^2) = \sum_{i=1}^n \frac{1}{n} \cdot (ci^2) \\ &= \frac{c}{n} \sum_{i=1}^n i^2 = \Theta(n^2) \end{aligned}$$



Randomized example 2

```
Func1( $A, n$ )  
  /*  $A$  is an array of integers  
1  $s \leftarrow 0$ ;  
2  $k \leftarrow \text{Random}(n)$ ;  
3 if  $k \leq \log n$  then  
4   | for  $i \leftarrow 1$  to  $n$  do  
5   |   |  $s \leftarrow s + A[i] * A[n]$ ;  
6   | end  
7 end  
8 return ( $s$ );
```



Running time analysis

- ▶ Worst case running time

- ▶ $T(n) = \Theta(n)$

- ▶ Best case?

- ▶ Expected analysis

- ▶ Step 1:

- ▶ Identify there are two cases: $k \leq \log n$ and $k > \log n$

- ▶ Step 2:

- ▶ Find probability of the two cases:

- $\Pr[k \leq \log n] = \frac{\log n}{n}; \Pr[k > \log n] = 1 - \frac{\log n}{n}$

- ▶ Step 3:

- ▶ Find time complexity for each case:

- $time(k \leq \log n) = cn; time(k > \log n) = c'$



Expected analysis

- ▶ Expected running time:

- ▶ $ET(n) = \Pr(k \leq \log n) \cdot \text{time}(k \leq \log n) + \Pr(k > \log n) \cdot \text{time}(k > \log n)$
 $= \frac{\log n}{n} \cdot (cn) + \left(1 - \frac{\log n}{n}\right) \cdot (c')$

- ▶ $\Rightarrow ET(n) = \Theta(\log n)$

These are artificial examples. We will see later a randomized algorithm for the sorting problem.



Part C: Lower bound theory



Problems and algorithms

- ▶ There can be many algorithms for solving the same problem.
 - ▶ Some have better time complexity than others.
- ▶ An important question:
 - ▶ For a given problem, what is the best possible time complexity?
- ▶ Such questions can be hard to answer
 - ▶ as typically we cannot “enumerate” all possible algorithms
- ▶ Often we try to provide a lower-bound
 - ▶ that is as tight as we can
 - ▶ Sometimes we know we have the right (tight) bound when there is an algorithm whose worst-case running time matches this lower bounds



Lower Bound

- ▶ No algorithm can have a faster (worst case) time complexity than a **theoretical lower bound**.

Definition:

$f(n)$ is a **theoretical lower bound** for a problem if every possible algorithm's worst-case time complexity is $\Omega(f(n))$.



A simple example

- ▶ The **Search** problem:

- ▶ Input: given an arbitrary array A of numbers and a key k
- ▶ Output: return whether $k \in A$ or not

- ▶ A trivial lower-bound

- ▶ $\Omega(1)$
- ▶ Not wrong, but useless

Can we get a better lower-bound?

- ▶ A better lower-bound

- ▶ $\Omega(n)$
- ▶ as in the worst case, any algorithm will have to inspect every element in A
- ▶



Tight Lower bound

- ▶ A lower-bound $f(n)$ for problem-P is **tight** if there exists an algorithm **A** for problem-P whose worst-case running time is $\Theta(f(n))$.
 - ▶ In some sense, the algorithm **A** has **optimal** running time.
- ▶ Back to the **Search** problem
 - ▶ There is an algorithm
 - ▶ $T(n) = \Theta(n)$
 - ▶ Hence the lower bound
 - ▶ $\Omega(n)$ is **tight**

```
def linear_search(A, k):  
    for i, x in enumerate(A):  
        if x == k:  
            return i  
    return None
```



FIN

